

CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

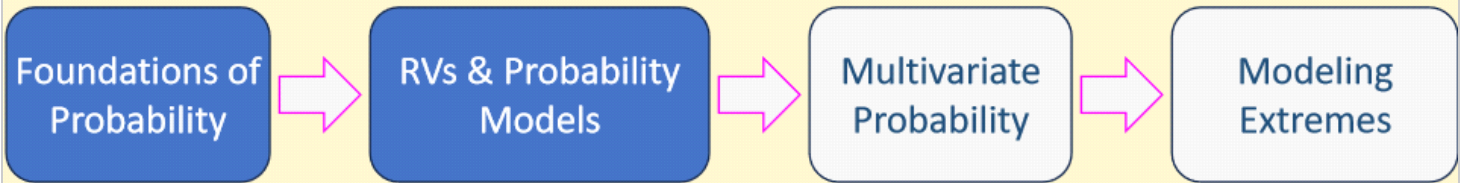
Dr. Prakash S Badal

Joint Probability Distributions

Ang and Tang (Section 3.3)

Course Flow

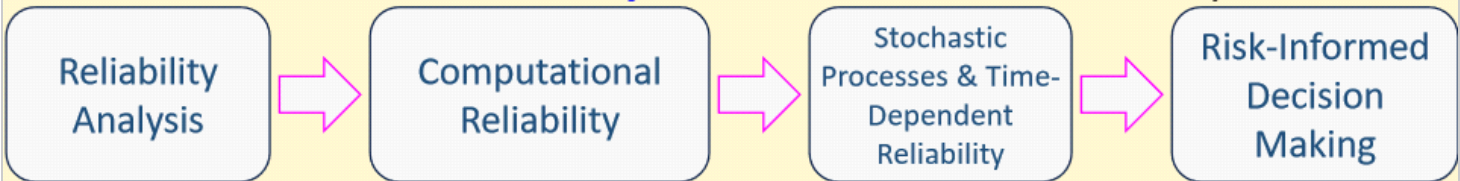
Module 1: **Probability** – Understanding Uncertainty



Module 2: **Statistics** – Learning from Data

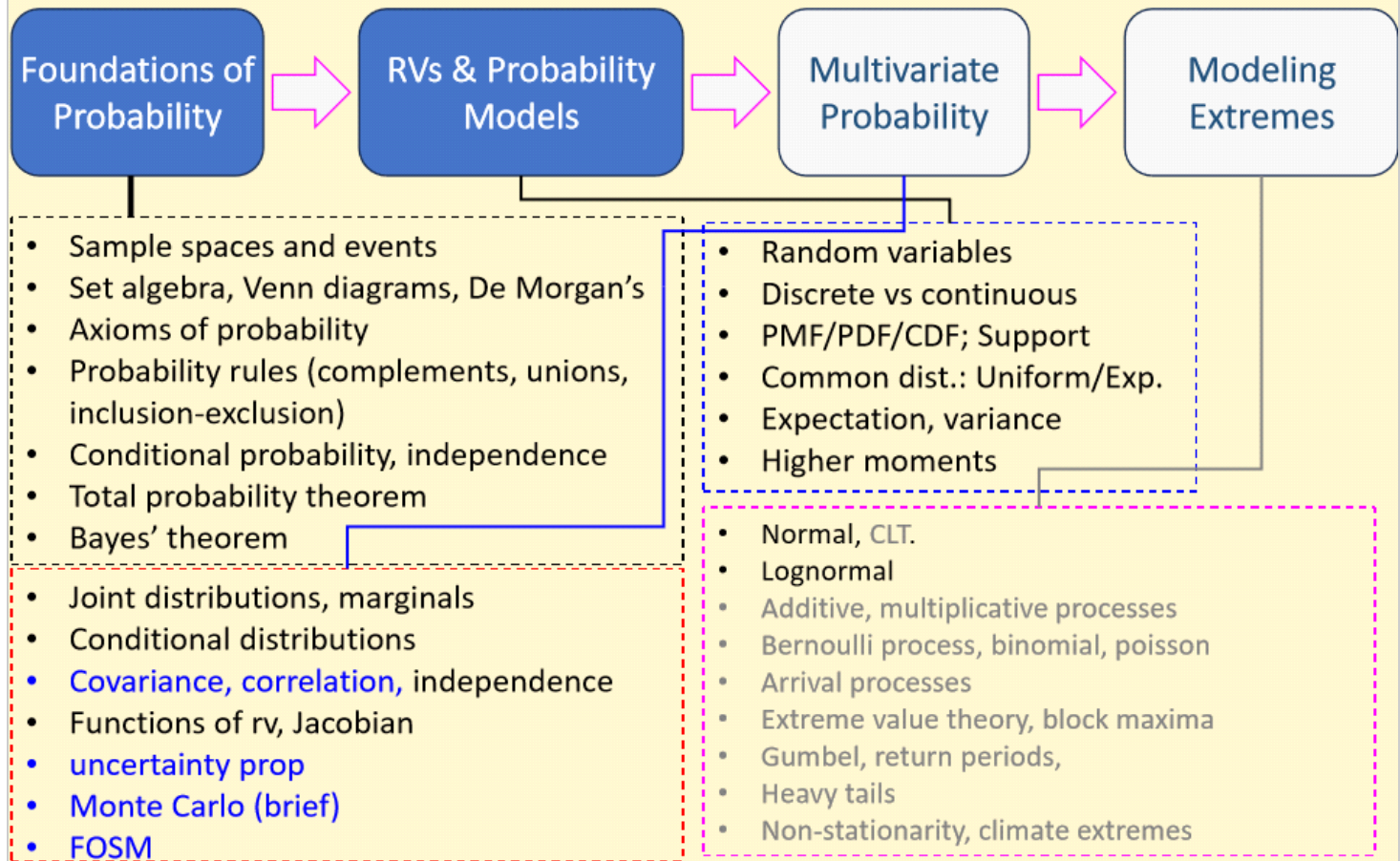


Module 3: **Reliability** – Decision under Uncertainty



Module I: Chapter 3 (M1C3)

Module 1: **Probability** – Understanding Uncertainty



L19
Sep 7, 2026

Messages from the Last Class

• Message 1: Joint RV

① Joint PDF

$$F_{XY}(x, y) = P_2(X \leq x, Y \leq y) = \int_{-\infty}^x \int_{-\infty}^y f_{XY}(u, v) du dv$$

② Joint PDF $\rightarrow$ Marginal PDFs

$$\int_{-\infty}^{\infty} f_{XY}(u, y) du dy \rightarrow f_Y(y)$$

Joint PDF $\rightarrow$ conditional PDFs

$$f_{Y|X}(y|X=x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

Joint probability distributions

P8. Independence

$$f_{XY}(x, y) \neq f_X(x) f_Y(y) \text{ in general}$$

If X & Y are statistically Independent

$$f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

P9. Conditional

$$f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$$

$$f_{Y|X}(y|x)$$

$$E[g(x)] = \int_{-\infty}^{\infty} g(x) \cdot f_X(x) dx$$

P10. Expected value:

$$g(x, y)$$

$$E[g(x, y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{XY}(x, y) dx dy$$

Practice Problem

When $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ and $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, their joint PDF is given by:

$$f_{XY}(x, y) =$$

$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp \left\{ -\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X} \right)^2 - 2\rho \left(\frac{x-\mu_X}{\sigma_X} \right) \left(\frac{y-\mu_Y}{\sigma_Y} \right) + \left(\frac{y-\mu_Y}{\sigma_Y} \right)^2 \right] \right\}$$

$-\infty < x, y < \infty; \sigma_X, \sigma_Y > 0; -1 < \rho < 1$

where ρ is the correlation coefficient between X and Y .

Find conditional PDF of Y given $X = x$.

→ PDF for $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} \exp \left\{ -\frac{1}{2} \left(\frac{x-\mu_X}{\sigma_X} \right)^2 \right\}$

Conditional PDF $f_{Y|X}(y|x) = \frac{f_{XY}(x, y)}{f_X(x)}$

$$\int_{-\infty}^{\infty} e^{-z^2/2} dz = \sqrt{2\pi}$$

Joint probability distributions

P11. Moments

$$M_{nk} = E[X^n Y^k] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^n y^k f_{xy}(x, y) dx dy$$

$$M_{00} = 1$$

$$M_{10} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{xy}(x, y) dx dy = \int_{-\infty}^{\infty} x f_x(x) dx = \mu_x$$

$$M_{01} = \mu_y$$

$$M_{11} = E[XY]$$

Joint probability distributions

p12. Central Moments

$$\eta_{nk} = E[(X - \mu_X)^n (Y - \mu_Y)^k]$$

$$\eta_{00} = 1$$

$$\eta_{10} = 0$$

$$\eta_{01} = 0$$

$$\eta_{20} = \sigma_X^2$$

$$\eta_{02} = \sigma_Y^2$$

$$\eta_{11} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)(y - \mu_Y) f_{XY}(x, y) dx dy$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (xy f_{XY} - x \mu_Y f_{XY} - y \mu_X f_{XY} + \mu_X \mu_Y f_{XY}) dx dy$$

$$= M_{11} - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y$$

$$= E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y]$$

p12a. COVARIANCE, $\text{COV}(X, Y) = E[XY] - E[X]E[Y]$
 $\xrightarrow{\text{No Squared}} \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)]$

Joint probability distributions

P13: Correlation Coefficient

$$\rho_{XY} = \frac{\text{COV}(X,Y)}{\sigma_X \sigma_Y} \equiv \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

$$\Rightarrow \text{COV}(X,Y) \equiv \sigma_{XY} = \rho_{XY} \sigma_X \sigma_Y$$

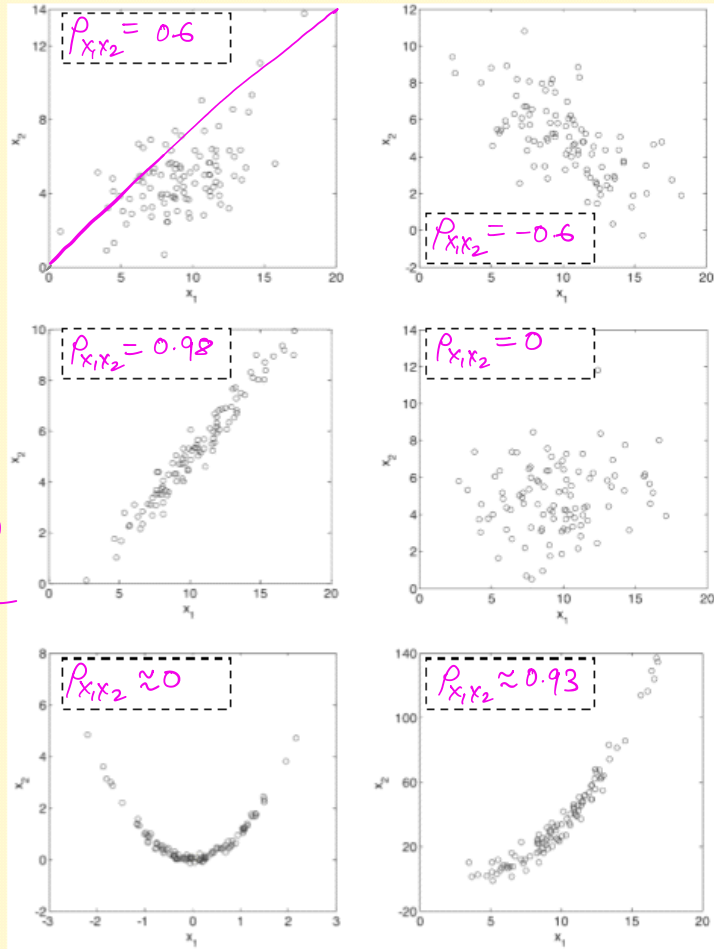
Correlation coefficient is a metric of predictability of "expense of one RV" given the "expense of another RV".

$\frac{X-\mu_X}{\sigma_X}$	$\frac{Y-\mu_Y}{\sigma_Y}$	
+1	+1	perfectly correlated $\rho_{XY} \rightarrow 1$
-0.5	-0.5	
-2.0	-2.0	
+1	-1	$\rho_{XY} \rightarrow -1$
-0.5	+0.5	
-2.0	+2.0	
+1	$\begin{matrix} +1 \\ -1 \end{matrix}$	$\rho_{XY} \rightarrow 0$
-0.5	$\begin{matrix} +0.5 \\ -0.5 \end{matrix}$	
-2.0	$\begin{matrix} +2.0 \\ -2.0 \end{matrix}$	

Correlation coefficient

ρ measures
linear dependence.

$\rho=0 \nRightarrow$
statistically Independence



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L20
Sep 8, 2026

Joint probability distributions

P14. Special case, Y is an affine transformation of X

$$Y = aX + b$$

$X: \mu_X, \sigma_X$ $Y: \mu_Y, \sigma_Y$

What is $\text{COV}[X, Y]$ and ρ_{XY} ?

Practice Problem

Total gravity load T on a floor is the sum of dead load D and live load L

$$T = D + L$$

Derive the general formula for the variance of T , i.e., σ_T^2 in terms of σ_D^2 , σ_L^2 , and σ_{DL} .

Example on covariance

If X is either 11 or 17; Y is either 20 or 8.

Find $\text{COV}[X, Y]$ and ρ_{XY} .

$$S = \{(11, 20), (11, 8), (17, 20), (17, 8)\}$$

(a) (11,20) and (17,8) have 0.5 probability each.

(b) All four with 0.25 probability each.

M1C3 lec-2

Exact Uncertainty Propagation

Given the distribution of the inputs, how to determine the uncertainty of the output exactly?

Engineering equations → Functions of multiple RVs → Linear transformations → Nonlinear transformations → Jacobian → Exact analytical transformation → Monte Carlo (briefly, as *numerical exact* method)

Message: Engineering never measures outputs directly.
it propagates uncertainty through models.

Starting of notes already
discussed in L13 (Aug 21,
2026)

Functions of Single rv

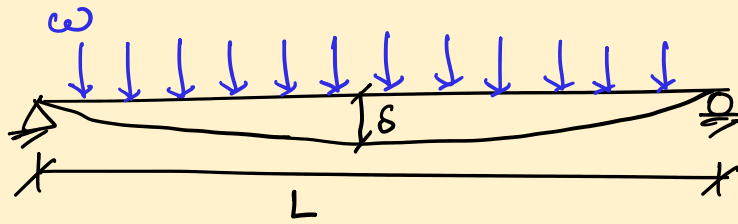
Ang A.H.S. and Tang (Section 4.2.1)

Why do we need functions of RV?

* wind speed $P_d = \frac{1}{2} \rho V^2$

$$V \longrightarrow P_d$$

* Beam



$$\delta = \frac{w L^4}{384 E I}$$

* Energy released in an Earthquake of magnitude M ,
 $E = 10^{1.5M}$

$$E_7/E_6 \approx 32$$

Why do we need functions of RV?

→ functions of 'observed RV' have physical significance.

$$g(x) = 2x + 3$$

$$E[g(x)] = 2 * E[x] + 3$$

$$\sigma^2[g(x)] = 2^2 * \sigma_x^2$$

$$\Rightarrow \sigma[g(x)] = 2 * \sigma_x$$

$$g(x) = x^2$$

$$E[g(x)] \stackrel{?}{=} g[E(x)]$$

$$g(x) = x^3 + 5 \sin x + e^x$$

$$E[g(x)] = E[x^3] + 5 E[\sin x] + E[e^x]$$

Expectation of Functions

Expectation is a linear operator

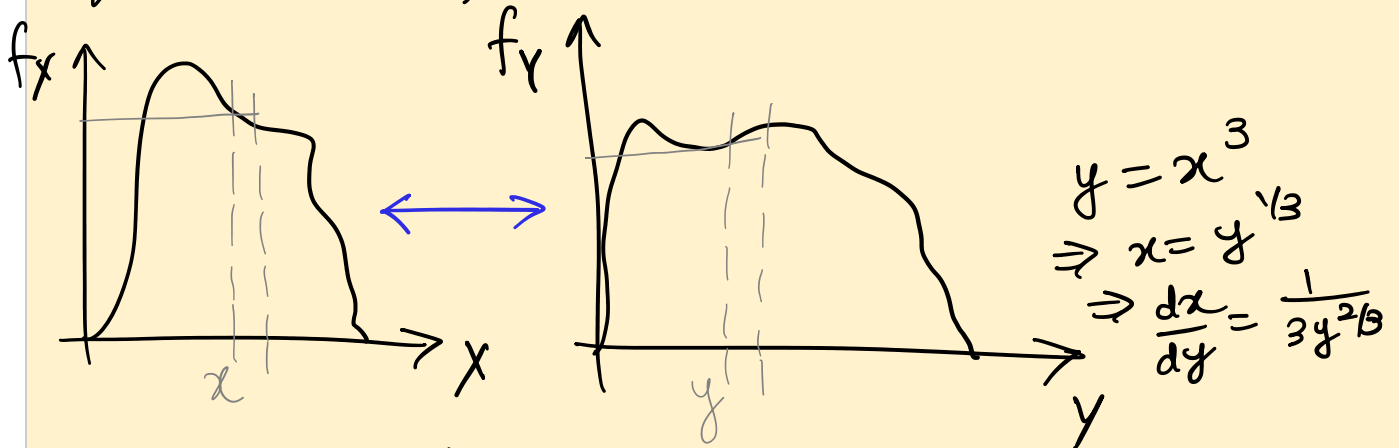
E & $g(x)$ are commutative, if $g(x)$ is
an affine transformation
 $ax+b$

$$E[g(x)] = g[E(x)]$$

Function of Single RV

Let's say, X has a pdf $f_X(x)$

find PDF for $Y = X^3$.



$$f_X dx = f_Y dy$$

$$\Rightarrow f_Y = f_X \left| \frac{dx}{dy} \right| = f_X (y^{1/3}) \cdot \left| \frac{1}{3y^{2/3}} \right|$$

Jacobian

M1C3 part-3/3

Approximate Uncertainty Propagation

What if the exact propagation is too complicated or impossible?

Exact propagation too difficult → Black-box engineering models → Taylor approximation
→ FOSM → Accuracy of approximation → Comparison with Monte Carlo

Message: Exactness is often too expensive.
Engineering relies on intelligent approximations.

Sources of uncertainties
Skip the sources example!

Sources of Uncertainty

- **Inherent uncertainty**: concrete strength; irreducible
- **Statistical uncertainty**: due to the lack of data; reducible
- **Characteristics of system**: material properties, dimensions
- **Demand** placed on the system
- **Mathematical models** used to analyze structure's behavior
- **Measurements uncertainty**: while assessing structure's health
- **Probabilistic models** used to describe uncertain quantities
- **Human error**

Categories of Uncertainty

In classical statistics,

- Aleatory
- Epistemic

Categories of Uncertainty

- Wind speed on the surface of a building:
 - Aleatory or epistemic?
- *Case 1: When there is no predictive wind model and only long-term statistical data is used?*
- *Case 2: When wind-fields are studied using precise computational fluid dynamics?*

Four Categories of Prediction

Nonlinear Functions of Single rv

Mean and Variance propagation laws

- If $Y = g(X)$ is a nonlinear function of X . Find $E[Y]$ & $\text{Var}(Y)$.

$$E[Y] = E[g(X)]$$

Taylor series expansion:

$$g(X) = g(\mu_X) + \left(\frac{\partial g}{\partial x}\right)_{\mu_X} (X - \mu_X) + \frac{1}{2!} \left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} (X - \mu_X)^2 + \text{H. O. T.}$$

$$\begin{aligned} E[Y] &\cong E\left(g(\mu_X) + \left(\frac{\partial g}{\partial x}\right)_{\mu_X} (X - \mu_X) + \frac{1}{2!} \left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} (X - \mu_X)^2\right) \\ &= g(\mu_X) + 0 + \frac{1}{2!} \left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} E[(X - \mu_X)^2] \end{aligned}$$

$$E[Y] \cong g(\mu_X)$$

First-order mean approximation

$$E[Y] \cong g(\mu_X) + \frac{1}{2} \left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} \sigma_X^2$$

Second-order mean approximation

Mean and Variance propagation laws

- If $Y = g(X)$ is a nonlinear function of X . Find $E[Y]$ & $\text{Var}(Y)$.

Taylor series expansion:

$$g(X) = g(\mu_X) + \left(\frac{\partial g}{\partial x}\right)_{\mu_X} (X - \mu_X) + \frac{1}{2!} \left(\frac{\partial^2 g}{\partial x^2}\right)_{\mu_X} (X - \mu_X)^2 + \text{H. O. T.}$$

$$\text{Var}[Y] = E[(Y - \mu_Y)^2] \cong \left[\left(\frac{\partial g}{\partial x}\right)_{\mu_X}\right]^2 \sigma_X^2 \quad \text{First-order var. approx.}$$

Example

- If $Y = \ln X$, what are expectation and variance of Y ? Use Taylor series approximations.

$$E[Y] \cong \ln \mu_X \quad \text{first order mean approximation}$$

$$E[Y] \cong \ln \mu_X - \frac{\delta_X^2}{2} \quad \text{second order mean approximation}$$

$$\text{Var}[Y] \cong \delta_X^2 \quad \text{first order variance approximation}$$

$$E[Y] \cong g(\mu_X) + \frac{1}{2} \left(\frac{\partial^2 g}{\partial x^2} \right)_{\mu_X} \sigma_X^2 \quad \text{Second-order mean approximation}$$

$$\text{Var}[Y] = E[(Y - \mu_Y)^2] \cong \left(\left(\frac{\partial g}{\partial x} \right)_{\mu_X} \right)^2 \sigma_X^2 \quad \text{First-order var. approx.}$$

Propagation Laws

Mean and Variance propagation laws

If $Y = a_1X_1 + a_2X_2 + c$, with a_i and c deterministic const.

$$E[Y] = a_1\mu_1 + a_2\mu_2 + c$$

$$\text{Var}[Y] =$$

Propagation: μ and σ

- Two structural engineers, Alice and Bob, independently measure the wind speed in m/s at the top of each tower.

Alice's reading $X \sim \mathcal{N}(\mu = 40, \sigma = 5)$

Bob's reading $Y \sim \mathcal{N}(\mu = 42, \sigma = 6)$

Their readings are correlated with $\rho = 0.8$

1. If Alice and Bob average their measurements, $W = (X + Y)/2$, what are $E[W]$ and $\text{Var}[W]$?
2. What's the covariance of X and Y ?
3. If design wind speed is 55 m/s, what's the probability that the average exceeds 55 m/s?



Propagation: mean and uncertainty

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Standard Normal Probabilities

Z-table

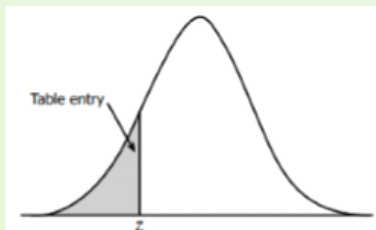


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Mean and Variance propagation laws

If $Y = a_1X_1 + a_2X_2 + \cdots + a_nX_n + c$, with a_i & c det. consts.

$$E[Y] = a_1X_1 + a_2X_2 + \cdots + a_nX_n + c$$

$$\text{Var}[Y] =$$

Monte Carlo Simulations (brief)

Monte Carlo Simulations



Stanisław Ulam



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Monte Carlo Simulations

Monte Carlo method [[edit](#)]

Late in the war, under the sponsorship of von Neumann, Frankel and Metropolis began to carry out calculations on the first general-purpose electronic computer, the [ENIAC](#) at the Aberdeen Proving Ground in Maryland. Shortly after returning to Los Alamos, Ulam participated in a review of results from these calculations.^[36] Earlier, while playing [solitaire](#) during his recovery from surgery, Ulam had thought about playing hundreds of games to estimate statistically the probability of a successful outcome.^[37] With ENIAC in mind, he realized that the availability of computers made such statistical methods very practical. John von Neumann immediately saw the significance of this insight. In March 1947 he proposed a statistical approach to the problem of neutron diffusion in fissionable material.^[38] Because Ulam had often mentioned his uncle, Michal Ulam, "who just had to go to Monte Carlo" to gamble, Metropolis dubbed the statistical approach "The [Monte Carlo method](#)".^[36] Metropolis and Ulam published the first unclassified paper on the Monte Carlo method in 1949.^[39]



Stanisław Ulam

John von Neumann

[Article](#) [Talk](#)

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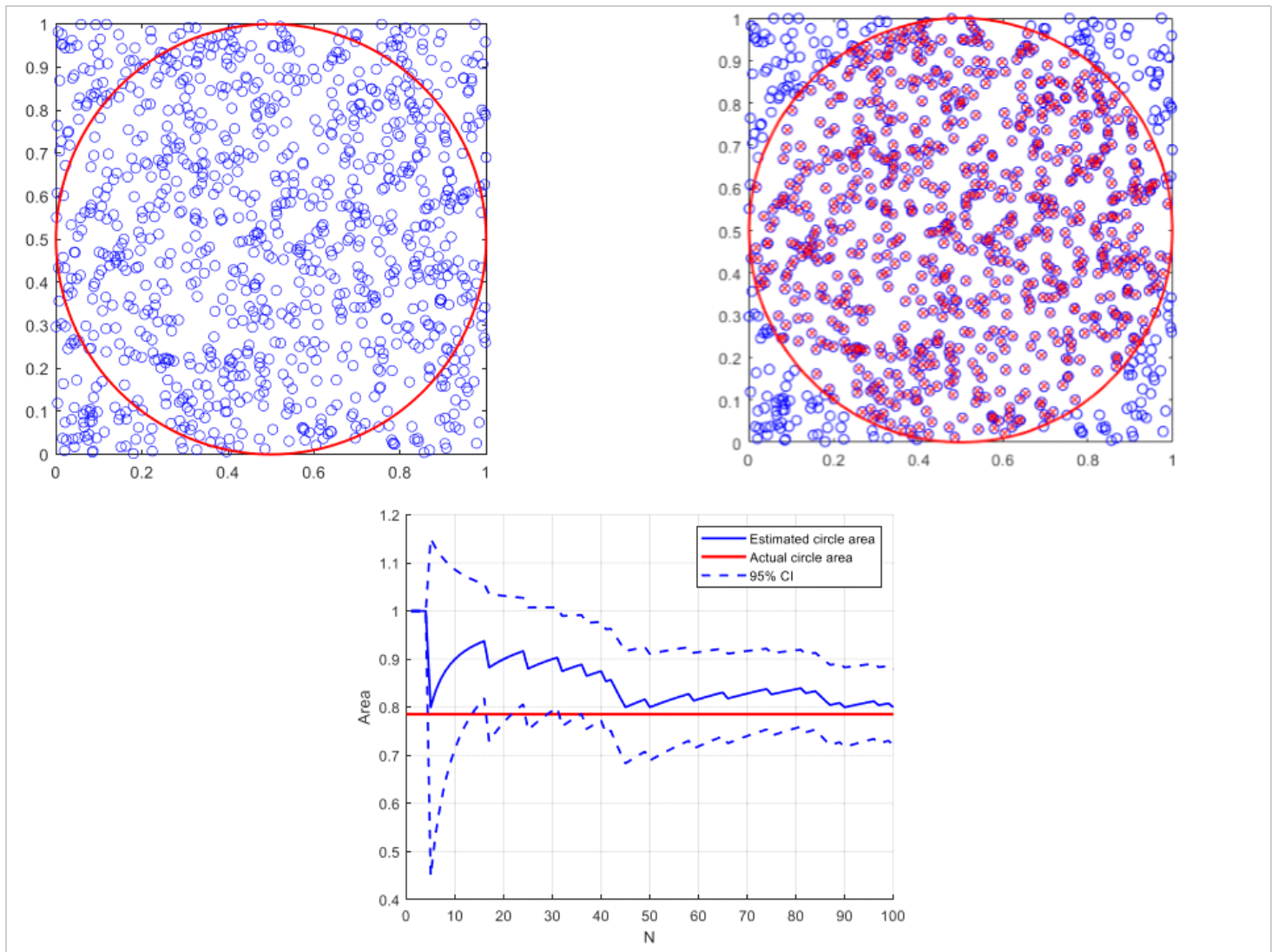
The native form of this personal name is Neumann János Lajos. This article uses Western name order when men.

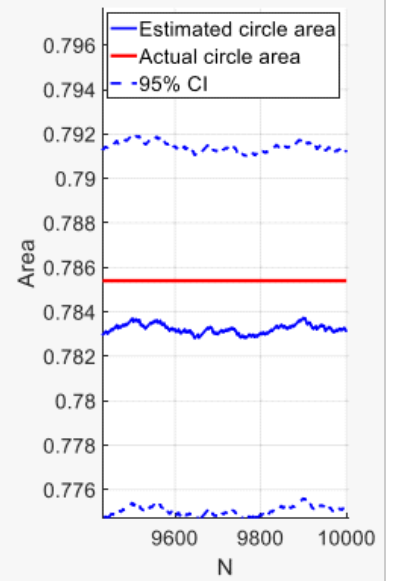
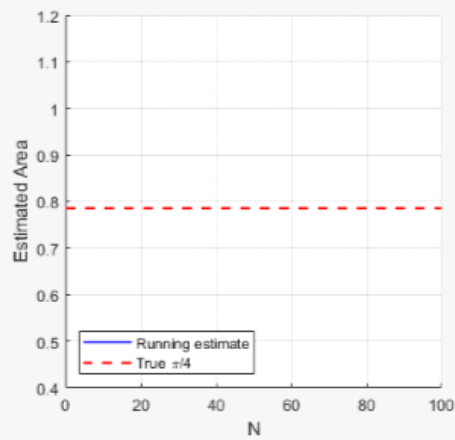
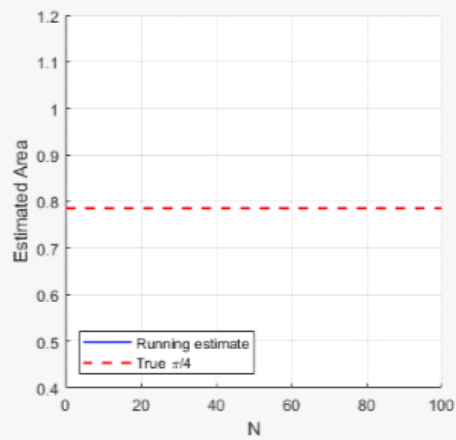
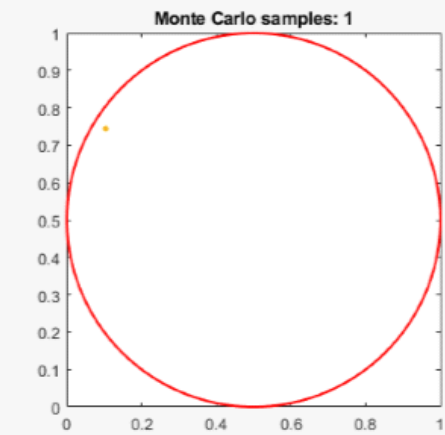
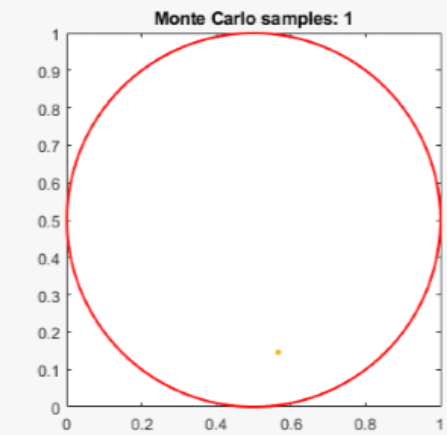
John von Neumann (/vɒn ˈnɔːmən/ *von NOY-men*; Hungarian: *Neumann János Lajos*;^[*note* 2] December 28, 1903 – February 8, 1957) was a Hungarian and American mathematician, physicist, computer scientist and engineer. Von Neumann had perhaps the widest coverage of any mathematician of his time,^[8] integrating [pure](#) and [applied](#) sciences and making [major contributions to many fields](#), including [mathematics](#), [physics](#), [economics](#), [computing](#), and [statistics](#). He was a pioneer in building the [mathematical framework of quantum physics](#), in the development of [functional analysis](#), and in [game theory](#), introducing or codifying concepts including [cellular automata](#), the [universal constructor](#) and the [digital computer](#). His analysis of the structure of [self-replication](#) [preceded the discovery of the structure of DNA](#).

During World War II, von Neumann worked on the [Manhattan Project](#). He developed the [mathematical models behind the explosive lenses](#) used in the [implosion-type nuclear weapon](#).^[9] Before and after the war, he consulted for



Circle area example in MATLAB





M1C3-Lec-4

Mon dd, 2026

M1C3 (part-4/3; discussion only)

Beyond correlation

Does correlation completely describe dependence?

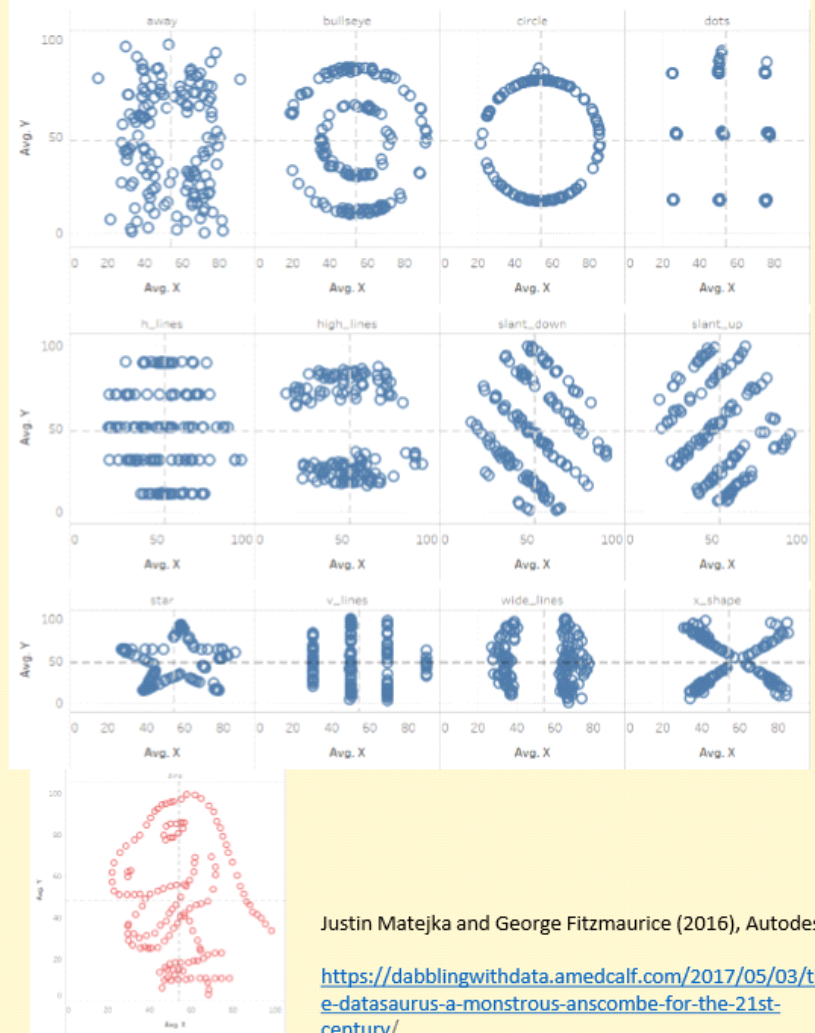
Review correlation → Examples where correlation fails → Tail dependence → Copulas → Joint hazard modelling (application to disasters)

Message: Correlation measures one aspect of dependence.

Nature is much richer.

Datasaurus Dozen

Datasaurus Dozen



Justin Matejka and George Fitzmaurice (2016), Autodesk

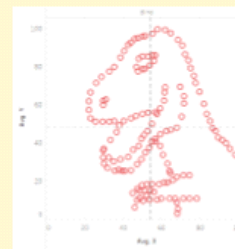
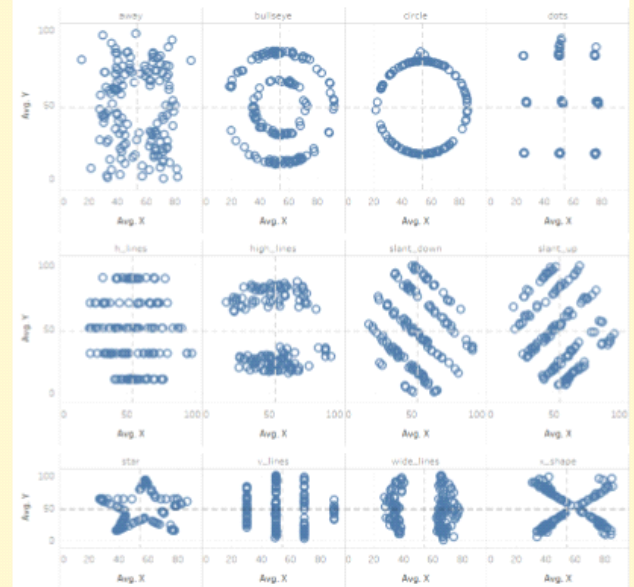
<https://dabblingwithdata.amedcalf.com/2017/05/03/the-datasaurus-a-monstrous-anscombe-for-the-21st-century/>

Datasaurus Dozen

<https://seanfobbe.com/tutorials/importance-of-visualization-datasaurus/>

	dataset	x_mean	x_sd	y_mean	y_sd	corr
	<char>	<num>	<num>	<num>	<num>	<num>
1:	dino	54.26327	16.76514	47.83225	26.93540	-0.06447185
2:	away	54.26610	16.76982	47.83472	26.93974	-0.06412835
3:	h_lines	54.26144	16.76590	47.83025	26.93988	-0.06171484
4:	v_lines	54.26993	16.76996	47.83699	26.93768	-0.06944557
5:	x_shape	54.26015	16.76996	47.83972	26.93000	-0.06558334
6:	star	54.26734	16.76896	47.83955	26.93027	-0.06296110
7:	high_lines	54.26881	16.76670	47.83545	26.94000	-0.06850422
8:	dots	54.26030	16.76774	47.83983	26.93019	-0.06034144
9:	circle	54.26732	16.76001	47.83772	26.93004	-0.06834336
10:	bullseye	54.26873	16.76924	47.83082	26.93573	-0.06858639
11:	slant_up	54.26588	16.76885	47.83150	26.93861	-0.06860921
12:	slant_down	54.26785	16.76676	47.83590	26.93610	-0.06897974
13:	wide_lines	54.26692	16.77000	47.83160	26.93790	-0.06657523

Datasaurus Dozen



Thank you!